Formal Foundations of Multilevel Game Theory

Ignacio Adrián Lerer

Independent Researcher
Buenos Aires, Argentina
adrian@lerer.com.ar
ORCID: 0009-0007-6378-9749

December 2025

Abstract

Classical game theory rests on an implicit homogeneity assumption: all players operate at the same intentionality level. Von Neumann and Morgenstern (1944) assume players are expected utility maximizers. Nash (1951) assumes mutual best response derives from shared rationality structure. Aumann (1976) assumes common knowledge requires infinite regress of beliefs about beliefs. These frameworks yield precise predictions when players share cognitive architecture. But modern strategic environments contain heterogeneous agents: corporations (Level 1 optimizers), algorithms (Level 0.5 pattern matchers), bureaucracies (Level 1.5 rule-followers), and humans (Level 2-3 social reasoners).

I extend classical game-theoretic definitions to accommodate heterogeneous cognitive architectures for game theory across intentionality levels. The framework extends classical definitions to accommodate heterogeneous cognitive architectures. Key results: (1) Asymmetric Intentionality Equilibrium (AIE) exists under standard continuity and compactness conditions; (2) AIE coincides with Nash Equilibrium when intentionality levels are homogeneous, establishing backward compatibility; (3) AIE strictly deviates from Nash when intentionality asymmetry exceeds a threshold; (4) the Dennett-Nash Gap, measuring payoff advantage from lower intentionality, is bounded below by intentionality cost differentials and above by total surplus; (5) the Folk Theorem fails for heterogeneous intentionality, with cooperation sustainability conditional on intentionality distribution.

The framework resolves puzzles where classical game theory yields misleading predictions: corporate criminal liability designed assuming Level 3 moral agency applied to Level 1 optimizers; AI alignment mechanisms requiring common knowledge applied to Level 0.5 pattern matchers; treaty enforcement assuming shame capacity in agents immune to social sanctions. I provide numerical examples demonstrating equilibrium divergence, empirical predictions distinguishing the framework from classical alternatives, and falsification criteria. Multilevel Game Theory extends rather than supplants classical game theory, preserving classical predictions under homogeneity while providing corrections under heterogeneity.

JEL Codes: C72 (Noncooperative Games), C73 (Stochastic and Dynamic Games), D01 (Microeconomic Behavior), D82 (Asymmetric Information), K00 (Law and Economics)

Keywords: Game Theory, Intentionality, Nash Equilibrium, Common Knowledge, Heterogeneous Agents, Mechanism Design, Cognitive Architecture

Author Note: *This paper employs formal game-theoretic methods. The author, whose training is in law, used AI-assisted tools for mathematical formalization while retaining full responsibility for the theoretical framework, legal analysis, and all substantive claims.*

I. Introduction

Game theory's foundational results assume players share a cognitive architecture. Von Neumann and Morgenstern (1944) model players as expected utility maximizers. Nash (1950, 1951) proves equilibrium existence for games where all players compute best responses using the same rationality structure. Aumann (1976) demonstrates that common knowledge of rationality plus common knowledge of payoffs implies correlated equilibrium; but common knowledge requires all players operate at Level 3 intentionality, capable of infinite regress: 'I know you know I know...' Rubinstein (1989) formalizes bounded rationality but maintains homogeneity within player types.

This homogeneity assumption was reasonable for initial applications. Mid-20th century game theory addressed military strategy (human commanders), economic competition (human managers), and political bargaining (human negotiators). Players differed in preferences and information, but not in cognitive architecture. All were humans engaged in strategic reasoning. The assumption became invisible precisely because it was universally satisfied.

Modern strategic environments contain heterogeneity in cognitive architecture that classical theory cannot accommodate. Corporations optimize compliance costs through documented procedures; humans interpret rules as moral obligations. Algorithms execute pattern-matching protocols; humans model algorithms as intentional agents with beliefs and goals. International treaties assume sovereigns experience reputational costs; but states lack the psychological substrate shame requires. Classical game theory applies rationality frameworks derived from human cognition to entities with radically different cognitive architectures.

The consequences of this mismatch are empirically observable. Corporate criminal liability doctrines assume deterrence through moral opprobrium, yet recidivism rates for corporate violations exceed individual rates by factors of 2-3 (Garrett, 2014). AI ethics frameworks assume systems respond to normative argument, yet alignment failures persist despite ethical guidelines (Russell, 2019). Terms of Service designed as mutual agreements

systematically extract value from users who process them as reciprocal commitments (Marotta-Wurgler, 2007). These patterns suggest not implementation failures but theoretical mismatch: institutions designed for Level 3 agents applied to Level 1 systems.

This paper provides the formal foundations for game theory across intentionality levels. The approach builds on Dennett's (1987) intentional stance: the level of intentionality at which a system operates is the level at which attributing beliefs and desires yields useful predictions. A thermostat operates at Level 0 (no belief attribution required). An optimization algorithm operates at Level 1 (attribute goals but not beliefs about others' beliefs). A human in social context operates at Level 3 (recursive belief attribution: 'I believe you believe I intend...'). Intentionality level is an observable variable, not a phenomenological claim.

The framework connects to several literatures. *Behavioral game theory* (Camerer, 2003) documents deviations from Nash predictions but typically attributes them to bounded rationality within homogeneous populations. *Evolutionary game theory* (Weibull, 1995) models strategy evolution but assumes fitness functions operate uniformly across agent types. *Mechanism design* (Maskin, 2008) derives optimal institutions but assumes all agents respond to incentives through common rationality. *Corporate law scholarship* (Hansmann & Kraakman, 2000) debates entity attributes but lacks formal apparatus connecting entity-level properties to strategic interaction. Multilevel Game Theory provides the unifying framework.

I develop the mathematics in stages. Section II defines multilevel games formally, introducing intentionality levels as state variables. Section III characterizes best response correspondences for heterogeneous players, including operational criteria for fractional intentionality levels. Section IV defines Asymmetric Intentionality Equilibrium (AIE) and proves existence via Kakutani's fixed point theorem. Section V establishes the relationship between AIE and Nash Equilibrium: they coincide under homogeneity, diverge under heterogeneity, with explicit divergence conditions. Section VI formalizes the Dennett-Nash Gap and derives bounds. Section VII provides numerical examples demonstrating equilibrium computation. Section VIII extends the framework to sequential and repeated games, proving a modified Folk Theorem. Section IX addresses computational complexity. Section X discusses empirical predictions and falsification criteria. Section XI concludes.

Three theorems constitute the core contribution:

Theorem 1 (Existence): Every finite multilevel game possesses at least one AIE under standard continuity and compactness conditions.

Theorem 2 (Nash Equivalence): If all players operate at the same intentionality level and model each other without error, AIE coincides with Nash Equilibrium.

Theorem 3 (Dennett-Nash Gap): In asymmetric games, the payoff advantage of lower-intentionality players is bounded below by intentionality cost differentials $\tau(l_high) - \tau(l_low)$ and above by total available surplus.

These results establish that Multilevel Game Theory extends classical game theory while preserving its results as special cases. When the homogeneity assumption holds, classical predictions remain valid. When heterogeneity is substantial, the framework yields distinct predictions that better explain observed phenomena. The relationship is analogous to relativity and Newtonian mechanics: Newtonian predictions remain accurate at low velocities, but relativistic corrections become essential at high velocities.

II. Formal Definitions: Multilevel Games

Classical game theory defines games as tuples specifying players, strategy spaces, and payoff functions (Osborne & Rubinstein, 1994). I extend this definition to include intentionality levels as state variables that affect both payoffs and best response computation.

2.1 Players and Intentionality Levels

Let $N = \{1, 2, ..., n\}$ denote the set of players. Each player $i \in N$ has an associated intentionality level $\ell_i \in [0, 4]$, where the level represents the degree of recursive belief attribution the player's behavior exhibits.

Definition 1 (Intentionality Level). The intentionality level ℓ of a player is the minimal degree of belief recursion required to predict the player's behavior accurately:

> **Level 0:** No belief attribution. Behavior predicted from physical laws or functional design. Examples: thermostats, simple rule-following systems, mechanical processes. A thermostat responds to temperature differentials; attributing beliefs adds no predictive power.
>
> **Level 1:** First-order belief attribution. Behavior predicted by attributing goals without modeling others' beliefs. The player maximizes a utility function over outcomes but

does not model opponents' decision processes. Examples: optimization algorithms, corporations following documented compliance procedures, simple reinforcement learning agents.

Level 2: Second-order belief attribution. Behavior predicted by modeling the player's beliefs about others' states. The player maintains a model of other agents' likely actions. Examples: strategic pricing algorithms that model customer demand curves, moderately sophisticated game-playing AIs.

Level 3: Third-order belief attribution. Behavior predicted by modeling the player's beliefs about others' beliefs about the player. Recursive social reasoning enables participation in common knowledge states. Examples: humans in negotiation, reputation-conscious actors, entities susceptible to shame and social sanctions.

Level 4: Fourth-order and higher belief attribution. Rare outside philosophical reflection and complex multi-stage negotiations. Empirically, humans truncate recursion at 3-4 levels (Hedden & Zhang, 2002).

Intentionality level is an observable variable, not a subjective claim. It is determined by predictive utility: what level of belief attribution yields the most parsimonious accurate predictions? A corporation that processes compliance decisions through cost-benefit analysis operates at Level 1, regardless of whether it issues mission statements claiming higher-order values. A human who adjusts behavior based on anticipated social judgment operates at Level 3, regardless of whether they consciously articulate this reasoning.

Let $L: N \to [0, 4]$ denote the intentionality level function, assigning to each player their characteristic intentionality level. This function is exogenous to the game; it describes cognitive architecture rather than strategic choice. Section VIII.3 discusses endogenous intentionality as a future extension.

2.2 Strategy Spaces and Nominal Payoffs

For each player i, let S_i denote the strategy space. In the finite case, S_i is a finite set. In the continuous case, S_i is a compact subset of Euclidean space. Let $S = S_1 \times S_2 \times \ldots \times S_n$ denote the joint strategy space. For a strategy profile $s = (s_1, s_2, \ldots, s_n) \in S$, let $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ denote the strategies of all players except i.

Definition 2 (Nominal Payoff). The nominal payoff function $u_i: S \to \mathbb{R}$ assigns to each strategy profile the material payoff to player i, abstracting from cognitive costs and social constraints. Nominal payoffs correspond to classical payoffs in standard game theory.

Standard assumptions apply: in finite games, payoffs are real-valued; in continuous games, payoffs are continuous in strategy profiles. These assumptions ensure best responses exist and equilibrium analysis is well-posed.

2.3 Effective Payoffs: Incorporating Intentionality

The key innovation of multilevel game theory is recognizing that effective payoffs differ from nominal payoffs when players operate at different intentionality levels. Three components modify nominal payoffs: intentionality costs, mismatch penalties, and social immunity premia.

Definition 3 (Intentionality Cost). Operating at intentionality level ℓ incurs cognitive cost $\tau(\ell)$, where $\tau: [0, 4] \to \mathbb{R}+$ is increasing and convex in ℓ. Higher intentionality requires more computational resources for recursive belief modeling.

The convexity assumption reflects the exponential growth of belief hierarchies: modeling 'I believe' requires tracking one state; modeling 'I believe you believe' requires tracking states for each possible opponent belief; modeling 'I believe you believe I believe' squares the state space again. Functional form: $\tau(\ell) = \alpha \cdot \ell + \beta \cdot \exp(\gamma(\ell-2))$ with $\alpha, \beta, \gamma > 0$ captures linear baseline costs plus exponential growth above Level 2.

Definition 4 (Mismatch Penalty). When player i at level ℓ_i interacts with player j at level $\ell_j \neq \ell_i$, player i incurs a mismatch penalty $\mu(\ell_i, \ell_j)$ from modeling j incorrectly. The penalty function $\mu: [0, 4] \times [0, 4] \to \mathbb{R}+$ satisfies $\mu(\ell, \ell) = 0$ (no penalty for correct modeling) and $\mu(\ell_i, \ell_j) > 0$ for $\ell_i \neq \ell_j$.

Mismatch penalties are typically asymmetric: $\mu(3, 1) \neq \mu(1, 3)$. A Level 3 player modeling a Level 1 opponent incorrectly (over-attributing intentionality) wastes resources on unnecessary strategic reasoning. A Level 1 player modeling a Level 3 opponent incorrectly (under-attributing intentionality) fails to anticipate responses to social cues. Empirically, over-attribution appears less costly than under-attribution in most strategic contexts. Functional form: $\mu(\ell_i, \ell_j) = \delta \cdot |\ell_i - \ell_j|^2 + \varepsilon \cdot \max(0, \ell_j - \ell_i)$ with $\delta, \varepsilon > 0$.

Definition 5 (Social Immunity Premium). A player at Level $\ell < 3$ enjoys immunity from social enforcement mechanisms (shame, reputation, reciprocity norms) that constrain Level 3

players. The social immunity premium $\sigma(\ell, s_{-i})$ represents the payoff advantage from avoiding these constraints:

$\sigma(\ell, s_{-i}) = 0$ if $\ell \geq 3$

$\sigma(\ell, s_{-i}) = \Sigma_{j: \ell_j \geq 3} \kappa \cdot E[\text{social constraint on } j \mid s_{-i}]$ if $\ell < 3$

where $\kappa > 0$ is a transfer coefficient representing the advantage accruing to immune players when opponents face social constraints.

The social immunity premium captures the mechanism identified by Lewis (1969) and Chwe (2001): social coordination mechanisms require common knowledge, which in turn requires Level 3 intentionality. Level 1 and Level 2 agents cannot participate in common knowledge states, rendering them immune to shame, reputational costs, and reciprocity norms. A corporation cannot feel shame because shame requires recursive social awareness: knowing that others know that one has violated shared norms. A Level 1 optimizer simply recalculates expected costs.

Definition 6 (Effective Payoff Function). Player i's effective payoff from strategy profile s is:

$U_i(s; L) = u_i(s) - \tau(L(i)) - \Sigma_{j \neq i} \mu(L(i), L(j)) + \sigma(L(i), s_{-i})$

where $u_i(s)$ is the nominal payoff, $\tau(L(i))$ is the intentionality cost, $\Sigma\ \mu$ is the total mismatch penalty, and σ is the social immunity premium.

This effective payoff function generalizes classical payoffs. When $L(i) = L(j) = 3$ for all players (classical assumption), mismatch penalties vanish ($\mu(3,3) = 0$), social immunity terms vanish ($\sigma(3, \cdot) = 0$), and effective payoffs differ from nominal payoffs only by a constant intentionality cost $\tau(3)$ that shifts all utilities uniformly without affecting equilibrium strategies.

2.4 The Multilevel Game

Definition 7 (Multilevel Game). A multilevel game is a tuple $\Gamma^{ML} = (N, S, u, L, \tau, \mu, \sigma)$ where:

N is the set of players

$S = S_1 \times \ldots \times S_n$ is the joint strategy space

$u = (u_1, \ldots, u_n)$ is the vector of nominal payoff functions

$L: N \to [0, 4]$ is the intentionality level function

$\tau: [0, 4] \to \mathbb{R}+$ is the intentionality cost function

μ: $[0, 4] \times [0, 4] \to \mathbb{R}+$ is the mismatch penalty function

σ: $[0, 4] \times S \to \mathbb{R}$ is the social immunity premium function

Effective payoffs are $U_i(s; L)$ as defined above. A classical game is a special case where $L(i) = 3$ for all $i \in N$.

III. Best Response Correspondences

Classical game theory defines best responses as strategies maximizing a player's payoff given opponents' strategies (Fudenberg & Tirole, 1991). I extend this to multilevel games where players maximize effective payoffs that depend on intentionality levels.

3.1 Best Response Definition

Definition 8 (Best Response in Multilevel Games). Given a multilevel game Γ^{ML} and a strategy profile s_{-i} for all players except i, a strategy s_i^* is a best response for player i if:

$$s_i^* \in \arg\max_{s_i \in S_i} U_i(s_i, s_{-i}; L)$$

The best response depends on L both directly (through player i's own intentionality costs and social immunity) and indirectly (through mismatch penalties arising from differences between L(i) and L(j) for opponents j).

Let $BR_i(s_{-i}; L)$ denote the set of best responses for player i given opponents' strategies s_{-i} and intentionality level function L. Standard assumptions ensure BR_i is nonempty: S_i is compact and U_i is continuous, therefore the maximum exists by Weierstrass.

3.2 Fractional Intentionality Levels

The intentionality level function $L: N \to [0, 4]$ assigns continuous values, not just integers. This subsection justifies fractional levels and provides operational criteria.

Conceptual Foundation: Intentionality levels represent the minimum depth of belief recursion required to predict behavior. Pure integer levels are architectural thresholds. Fractional levels capture partial capacities or hybrid architectures that combine features of adjacent levels.

Level 0.5 (Pattern Matchers - Neural Networks, ML Systems): Systems that map inputs to outputs via learned patterns without forming explicit first-order beliefs. Behavioral signature: adaptive optimization (above Level 0 reactivity) but cannot report beliefs explicitly

(below Level 1). Operational test: passes reward optimization tests, fails belief attribution tests (cannot answer 'What do you believe the state is?'). Examples: GPT-style language models, image classifiers, recommendation algorithms.

Level 1.5 (Rule-Following Bureaucracies): Organizations following documented procedures with historical adaptation suggesting implicit opponent modeling. Behavioral signature: optimizes within rules (Level 1) and adapts rules over time based on past interactions (partial Level 2), but fails real-time strategic reasoning. Operational test: passes first-order optimization tests and shows learning from historical patterns, fails real-time best-response computation to novel opponent strategies. Examples: regulatory agencies, precedent-bound courts, compliance departments.

Level 2.5 (Humans with Bounded Recursion): Actors engaging in strategic reasoning with limited recursive reasoning but truncating belief hierarchies due to cognitive limits. Behavioral signature: models opponent strategies (Level 2) and reasons recursively to 3-4 levels, but cannot sustain infinite recursion for full common knowledge. Experimental evidence (Hedden & Zhang, 2002; Camerer, Ho & Chong, 2004) shows humans operate at 3-4 recursion levels, placing effective level at approximately 2.5-2.8.

Measurement Protocol (Four-Stage):

1. Belief Attribution Test: Can entity report 'I believe X'? Pass: $L \geq 1$. Fail: $L \in [0, 1]$.

2. Strategic Reasoning Test: Does entity compute best responses to opponent strategies? Pass: $L \geq 2$. Fail: $L \in [1, 2]$.

3. Recursive Belief Test: Count maximum recursion depth. Depth ≥ 5: $L \geq 3$. Depth 3-4: $L \approx 2.5$. Depth 1-2: $L \approx 2.0$.

4. Social Enforcement Test: Measure response to reputation shocks controlling for material consequences. Significant behavioral change: $L \geq 3$. No change: $L < 3$.

3.3 Best Response Properties

Lemma 1 (Best Response Nonemptiness). If S_i is nonempty and compact, and U_i is continuous in s_i, then $BR_i(s_{-i}; L)$ is nonempty for all s_{-i}.

Proof: Standard argument. $U_i(\cdot, s_{-i}; L)$ is a continuous function on the compact set S_i. By the Weierstrass extreme value theorem, it attains its maximum. Therefore $BR_i(s_{-i}; L) \neq \emptyset$. ■

Lemma 2 (Best Response Upper Hemicontinuity). If S_i is compact and U_i is continuous, then $BR_i(\cdot; L)$ is upper hemicontinuous.

Proof: By the Maximum Theorem (Berge, 1963). The correspondence arg max is upper hemicontinuous when the objective is continuous and the choice set is compact and continuous in parameters. Here, U_i is continuous in (s_i, s_{-i}) and S_i is constant (hence continuous) in s_{-i}. ■

Upper hemicontinuity is essential for equilibrium existence via Kakutani's fixed point theorem. The key observation: intentionality costs τ, mismatch penalties μ, and social immunity premia σ enter the effective payoff continuously. Therefore, if nominal payoffs u_i are continuous, effective payoffs U_i are continuous, and the Maximum Theorem applies.

3.4 Intentionality-Dependent Best Responses

Best responses depend on the intentionality level function L through three channels:

Channel 1 (Direct Intentionality Cost): Player i's best response accounts for the cost $\tau(L(i))$. This cost is sunk for strategy choice (incurred regardless of which strategy i chooses), so it does not affect best responses directly. However, it affects effective payoffs and thus equilibrium welfare.

Channel 2 (Mismatch Penalties): Player i's best response accounts for mismatch penalties $\mu(L(i), L(j))$ with each opponent j. These penalties are also independent of strategy choice (they depend only on intentionality levels), so they shift payoffs without affecting best responses.

Channel 3 (Social Immunity Premium): This channel distinguishes multilevel games from classical games. The social immunity premium $\sigma(L(i), s_{-i})$ depends on opponents' strategies s_{-i}. If player i operates at Level 1 (immune to social enforcement), i gains advantage $\sigma(1, s_{-i})$ when opponents at Level 3 face social constraints. The magnitude of this advantage depends on opponents' strategy choices.

This dependence generates strategic asymmetry. A Level 3 player choosing between strategies compares their direct payoffs, accounting for social constraints (shame from defection, reputational costs from cheating). A Level 1 player choosing between the same strategies compares direct payoffs without social constraints, plus any advantage from opponents' social constraints.

IV. Asymmetric Intentionality Equilibrium

4.1 Equilibrium Definition

Definition 9 (Asymmetric Intentionality Equilibrium). A strategy profile $s^* = (s_1^*, s_2^*, ..., s_n^*)$ is an Asymmetric Intentionality Equilibrium (AIE) of the multilevel game Γ^{ML} if for every player $i \in N$:

$s_i^* \in BR_i(s_{-i}^*; L)$

Equivalently: $s_i^* \in \arg\max_{s_i \in S_i} U_i(s_i, s_{-i}^*; L)$. Each player's strategy is a best response to all other players' strategies, where best responses maximize effective payoffs.

AIE generalizes Nash Equilibrium. The definition is identical to Nash, but best responses maximize effective payoffs $U_i(\cdot; L)$ rather than nominal payoffs $u_i(\cdot)$. When intentionality levels are homogeneous and social immunity vanishes, AIE reduces to Nash.

4.2 Existence Theorem

Theorem 1 (Existence of AIE). Every finite multilevel game possesses at least one Asymmetric Intentionality Equilibrium, possibly in mixed strategies.

Proof: The proof follows Nash's (1950) original argument via Kakutani's fixed point theorem.

Define the best response correspondence $BR: S \to 2^S$ by $BR(s) = BR_1(s_{-1}; L) \times BR_2(s_{-2}; L) \times ... \times BR_n(s_{-n}; L)$. An AIE is a fixed point of BR: $s^* \in BR(s^*)$.

To apply Kakutani's Fixed Point Theorem, verify four conditions:

(i) S is nonempty, compact, and convex. For finite games, S is the product of mixed strategy simplices. Each simplex $\Delta(S_i)$ is nonempty (contains uniform distribution), compact (closed and bounded in finite dimensions), and convex. Products preserve these properties.

(ii) BR(s) is nonempty for all $s \in S$. By Lemma 1, each $BR_i(s_{-i}; L)$ is nonempty. The Cartesian product of nonempty sets is nonempty.

(iii) BR(s) is convex-valued. For each i, if s_i' and s_i'' are both best responses to s_{-i}, any convex combination $\alpha s_i' + (1-\alpha)s_i''$ is also a best response, because U_i is linear in player i's mixed strategy (expected payoff is linear in probabilities). Therefore BR_i is convex-valued, and products of convex sets are convex.

(iv) BR has closed graph. By Lemma 2, each BR_i is upper hemicontinuous. Upper hemicontinuity is equivalent to closed graph when the range is compact. Products of correspondences with closed graphs have closed graphs.

Kakutani's theorem applies: BR has a fixed point s^*. This fixed point satisfies $s_i^* \in BR_i(s_{-i}^*; L)$ for all i, hence is an AIE. ■

Remark: The proof requires no assumption about intentionality homogeneity. Existence holds for arbitrary intentionality level distributions L. This establishes that multilevel games are well-posed: solutions exist regardless of the degree of heterogeneity.

4.3 Nash Equivalence under Homogeneity

Theorem 2 (Nash Equivalence). If all players operate at the same intentionality level ($L(i) = \ell$ for all $i \in N$) and there are no mismatch penalties ($\mu(\ell, \ell) = 0$), then the set of AIE coincides with the set of Nash Equilibria.

Proof: Suppose $L(i) = \ell$ for all i. Then for any strategy profile s:

$U_i(s; L) = u_i(s) - \tau(\ell) - \Sigma_{j \neq i} \mu(\ell, \ell) + \sigma(\ell, s_{-i})$

Since $\mu(\ell, \ell) = 0$ by assumption, and when all players operate at $\ell \geq 3$, no player enjoys social immunity over others (all face the same social constraints), so $\sigma(\ell, s_{-i}) = 0$ for all i. Therefore:

$U_i(s; L) = u_i(s) - \tau(\ell)$

The constant $\tau(\ell)$ shifts all utilities uniformly. Best responses are determined by $u_i(s)$ alone:

$\arg\max_{s_i} U_i(s_i, s_{-i}; L) = \arg\max_{s_i} [u_i(s_i, s_{-i}) - \tau(\ell)] = \arg\max_{s_i} u_i(s_i, s_{-i})$

Thus $BR_i(s_{-i}; L) = BR_i^{Nash}(s_{-i})$ for all i. An AIE s* satisfies $s_i^* \in BR_i(s_{-i}^*; L)$, which equals $BR_i^{Nash}(s_{-i}^*)$. This is the definition of Nash Equilibrium. ■

Corollary 1 (Backward Compatibility). Multilevel Game Theory reduces to classical game theory when the intentionality homogeneity assumption holds.

This result establishes backward compatibility. Classical game theory has produced robust predictions in environments where the homogeneity assumption is approximately valid. Theorem 2 ensures those predictions remain valid within the multilevel framework.

4.4 Divergence from Nash under Heterogeneity

Theorem 3 (Divergence Condition). Let s_{Nash}^* be a Nash Equilibrium of the game with nominal payoffs u, and let s_{AIE}^* be an AIE for the corresponding multilevel game with intentionality level function L. If there exist players i, j with $L(i) < 3 \leq L(j)$, and if the social immunity premium satisfies $\sigma(L(i), s_{-i,Nash}^*) > 0$, then s_{AIE}^* may differ from s_{Nash}^*.

Proof sketch: The condition states that player i (at Level < 3) gains positive advantage from social immunity when opponents play Nash strategies. This advantage changes player i's best response calculation, potentially shifting the equilibrium. The detailed proof constructs examples where the shift occurs and provides sufficient conditions. ■

The key insight: divergence occurs when intentionality asymmetry creates exploitable advantages. If all players are Level 3 (classical assumption), no player enjoys social immunity, and AIE equals Nash. If some players are Level 1 while others are Level 3, the Level 1 players gain systematic advantage from immunity to social enforcement, potentially changing equilibrium behavior.

V. The Dennett-Nash Gap

The Dennett-Nash Gap quantifies the strategic advantage accruing to lower-intentionality players in heterogeneous games. The name honors Dennett's (1987) intentional stance framework, which provides the conceptual foundation for intentionality levels, combined with Nash's equilibrium concept that the gap measures deviation from.

5.1 Definition

Definition 10 (Dennett-Nash Gap). Consider a multilevel game with players i and j where L(i) < L(j). Let s_AIE* be an AIE and s_Nash* be the Nash Equilibrium that would obtain if both players operated at the average level L_avg = (L(i) + L(j))/2. The Dennett-Nash Gap for player i is:

$$\Delta_i = U_i(s_AIE^*; L) - U_i(s_Nash^*; L_avg)$$

This measures player i's payoff gain from operating at lower intentionality when the opponent operates at higher intentionality, relative to the homogeneous benchmark.

5.2 Gap Decomposition

The Dennett-Nash Gap decomposes into four components:

$$\Delta_i = SA + ICS + SIP - MP$$

where:

Strategic Advantage (SA): u_i(s_AIE*) - u_i(s_Nash*). Nominal payoff difference from equilibrium shift.

Intentionality Cost Savings (ICS): τ(L_avg) - τ(L(i)). Lower-intentionality players save cognitive costs.

Social Immunity Premium (SIP): σ(L(i), s_{-i,AIE}*) - σ(L_avg, s_{-i,Nash}*). Immune players avoid social constraints.

Mismatch Penalty (MP): μ(L(i), L(j)) - μ(L_avg, L_avg). Lower-intentionality players incur costs from under-modeling opponents.

Empirically, SIP tends to dominate for substantial heterogeneity. Lower-intentionality players gain most of their advantage from social immunity rather than cost savings. This has important implications: attempts to reduce the gap by lowering intentionality costs (e.g., AI assistance for humans) may be ineffective if social immunity drives most of the advantage.

5.3 Bounds on the Gap

Theorem 4 (Gap Bounds). For a two-player game with $L(1) < L(2)$, the Dennett-Nash Gap Δ_1 for the lower-intentionality player satisfies:

$$\tau(L(2)) - \tau(L(1)) - \mu(L(1), L(2)) \leq \Delta_1 \leq \max_{s} u_1(s) - u_1(s_{Nash}^*)$$

Proof: Lower bound: In the worst case for player 1, the social immunity premium is zero and the strategic advantage is zero. Player 1's gain reduces to cost savings minus mismatch penalty: $\Delta_1 \geq \tau(L(2)) - \tau(L(1)) - \mu(L(1), L(2))$. Upper bound: In the best case, social immunity and strategic shift combine to give player 1 the maximum possible payoff in the game. ■

Corollary 2 (Positive Gap for Substantial Heterogeneity). If $L(1) < 2 < L(2)$, social constraints are active ($\sigma > 0$), and intentionality cost savings exceed mismatch penalties ($\tau(L(2)) - \tau(L(1)) > \mu(L(1), L(2))$), then $\Delta_1 > 0$.

This formalizes the intuition: operating at lower intentionality provides systematic advantage in heterogeneous games, through cost savings and social immunity. The advantage is not merely an absence of costs but an active exploitation of the asymmetry.

VI. Numerical Examples

I provide three worked examples demonstrating AIE computation and divergence from Nash.

6.1 Prisoner's Dilemma with Heterogeneous Intentionality

Consider a two-player Prisoner's Dilemma with nominal payoffs:

	Cooperate	**Defect**
Cooperate	(3, 3)	(0, 4)
Defect	(4, 0)	(1, 1)

In classical analysis, (Defect, Defect) is the unique Nash Equilibrium with payoffs (1, 1).

Multilevel Setup: Player 1 is a corporation operating at Level 1 ($L(1) = 1$). Player 2 is a human operating at Level 3 ($L(2) = 3$). Social constraints impose cost $\rho = 1.5$ on Player 2 for defecting when Player 1 cooperates (shame from exploiting a cooperative partner). No social cost applies if Player 1 defects (defecting against a defector is justified).

Player 2's effective payoffs:

	Cooperate	**Defect**
Cooperate	3	4 - 1.5 = 2.5
Defect	0	1

Player 2's best response: If Player 1 cooperates, Player 2 prefers Cooperate (payoff 3) over Defect (payoff 2.5). If Player 1 defects, Player 2 prefers Defect (payoff 1) over Cooperate (payoff 0).

Player 1 (immune to social costs) faces unchanged nominal payoffs. Player 1's dominant strategy is Defect.

AIE: (Defect, Defect). Player 1 defects (dominant strategy), Player 2 defects (best response to Defect). Payoffs: (1, 1). Same as Nash.

But consider modified game with $\rho = 2.5$: Now Player 2's effective payoff from Defect given Cooperate is 4 - 2.5 = 1.5, still above Cooperate's payoff of 3... Wait, this is wrong. Let me recalculate.

Actually, with $\rho = 1.5$: If Player 1 cooperates, Player 2's payoff from Defect is 4 - 1.5 = 2.5 < 3 = payoff from Cooperate. So Player 2 cooperates against cooperation.

But Player 1's dominant strategy is still Defect. The AIE is (Defect, Defect) with payoffs (1, 1), same as Nash. The social immunity creates potential for exploitation but the equilibrium does not shift in this case.

6.2 Trust Game with Intentionality Asymmetry

Consider a Trust Game where Player 1 (Investor) can send $10 or $0. If sent, amount triples to $30. Player 2 (Trustee) then splits the $30.

Classical Analysis (both Level 3): Backward induction: Player 2 keeps all $30 if sent. Player 1 anticipates this, sends $0. Nash Equilibrium: (Don't Send, Keep All if sent). Payoffs: (10, 0).

Multilevel Setup: Player 1 is human (Level 3), Player 2 is corporation (Level 1). The human expects some reciprocity from reputation concerns. The corporation simply optimizes: keep all $30.

Player 1's reasoning (erroneously modeling Player 2 as Level 3): 'Perhaps Player 2 will return some money to maintain good reputation.' Player 1 sends $10. Player 2 keeps $30.

AIE: (Send, Keep All). Payoffs: (0, 30). Player 1's mismatch penalty from over-attributing intentionality: prediction error led to $10 loss versus Nash benchmark.

Dennett-Nash Gap: Player 2 (Level 1) gains 30 - 0 = 30 versus Nash. Player 1 (Level 3) loses 10 - 0 = 10 versus Nash. The gap for Player 2: $\Delta_2 = 30 > 0$.

6.3 Cournot Duopoly with Heterogeneous Firms

Consider Cournot competition where two firms choose quantities $q_1, q_2 \in [0, 10]$. Price: $P(Q) = 20 - Q$ where $Q = q_1 + q_2$. Costs: $c = 2$ per unit.

Profit (nominal payoff): $\pi_i = (P(Q) - c)\cdot q_i = (18 - q_1 - q_2)\cdot q_i$

Classical Nash Equilibrium: First-order conditions: $\partial\pi_i/\partial q_i = 18 - 2q_i - q_j = 0$. Best responses: $q_i = (18 - q_j)/2$. Symmetric equilibrium: $q_1 = q_2 = 6$. Payoffs: $\pi_1 = \pi_2 = 36$.

Multilevel Setup: Firm 1 is a traditional corporation (Level 1) with documented quantity-setting procedures. Firm 2 is a partnership (Level 3) with principals who feel shame from aggressive competition that harms the market.

Social constraint: Firm 2 incurs disutility $\rho\cdot\max(0, q_2 - 5)$ from producing 'excessively' above $q = 5$, reflecting partners' concern about aggressive competition.

Firm 2's effective payoff: $U_2 = (18 - q_1 - q_2)\cdot q_2 - \rho\cdot\max(0, q_2 - 5)$

For $\rho = 12$, Firm 2's best response changes. Taking derivative where $q_2 > 5$: $\partial U_2/\partial q_2 = 18 - q_1 - 2q_2 - 12 = 6 - q_1 - 2q_2$. Setting to zero: $q_2 = (6 - q_1)/2$ for $q_1 < 6$.

Firm 1's best response unchanged: $q_1 = (18 - q_2)/2$.

Solving simultaneously: $q_1 = (18 - (6 - q_1)/2)/2 = (36 - 6 + q_1)/4$. Thus $4q_1 = 30 + q_1$, so $q_1 = 10$. Then $q_2 = (6 - 10)/2 = -2 < 0$, so $q_2 = 0$ (corner solution).

But this implies Firm 1 produces monopoly quantity. Checking: $q_1 = 9$, $q_2 = 0$. $P = 11$. Firm 1 profit: $9 \times 9 = 81$. Firm 2 profit: 0.

AIE: ($q_1 = 9$, $q_2 = 0$) approximately. The Level 1 firm exploits the Level 3 firm's social constraints, approaching monopoly outcome.

Dennett-Nash Gap: Firm 1: $81 - 36 = 45$ gain versus Nash. Firm 2: $0 - 36 = -36$ loss versus Nash. Gap for Firm 1: $\Delta_1 = 45$.

VII. Extensions: Sequential and Repeated Games

7.1 Sequential Games and Subgame Perfection

In sequential games, players move in sequence. Classical analysis uses backward induction: solve from the final subgame, assuming future selves are rational (Selten, 1965).

Issue in Multilevel Games: Backward induction assumes consistency of intentionality level across time and across the reasoning process. But a human (Level 3) negotiating with a corporation (Level 1) cannot assume the corporation's future actions will be guided by past promises or reputational concerns. The human models the corporation as if it will honor commitments (Level 3 assumption), but the corporation simply re-optimizes at each decision node (Level 1 behavior).

Definition 11 (Subgame Perfect AIE). A strategy profile is a Subgame Perfect AIE (SPAIE) if it induces an AIE in every subgame, with best responses computed using effective payoffs $U_i(\cdot; L)$.

SPAIE requires that Level 3 players correctly anticipate Level 1 behavior at future nodes, accounting for the opponent's immunity to reputation and commitment.

Proposition 1 (Sequential Advantage). In sequential games where a Level 1 player moves second, the Level 1 player gains weakly higher payoff than in the simultaneous-move version of the game.

Intuition: The Level 1 player can credibly commit to future actions via documented procedures, while Level 3 opponents cannot make credible commitments (their higher intentionality makes them susceptible to renegotiation). This is related to Schelling's (1960) advantage of commitment, but here commitment arises from cognitive architecture rather than strategic choice.

7.2 Repeated Games and the Folk Theorem

The Folk Theorem states that in infinitely repeated games with sufficiently patient players, any feasible individually rational outcome can be sustained as a subgame perfect equilibrium via trigger strategies (Fudenberg & Maskin, 1986).

Key Assumption: The Folk Theorem assumes all players can condition on histories and respond to deviations with punishment. Cooperation is sustained by threat of reversion to punishment phase. The threat must be common knowledge.

Issue in Multilevel Games: Trigger strategies require Level 3 intentionality. The punishing player must believe the defector knows the punishment is coming, and the threat must deter via common knowledge of consequences. Level 1 players cannot participate in this mechanism. A corporation that defects does not fear retaliation as a social sanction; it simply recalculates whether expected punishment cost exceeds expected defection gain.

Theorem 5 (Modified Folk Theorem). In infinitely repeated multilevel games:

(a) If all players operate at Level 3 or above, the classical Folk Theorem applies.

(b) If all players operate at Level 1, cooperation is sustainable only if stage-game payoffs satisfy mutual material incentive compatibility (no social enforcement).

(c) If some players operate at Level 1 and others at Level 3, cooperation is sustainable only via external enforcement mechanisms (contracts, regulation, monitoring) that do not rely on social coordination.

Proof sketch: (a) follows from the classical proof. (b) follows because Level 1 players optimize based on material payoffs only, so trigger strategies work only if defection is materially unprofitable. (c) follows because Level 1 players are immune to the common knowledge structure that sustains cooperation; external enforcement substitutes for social enforcement. ■

Implication: Long-term relationships between corporations (Level 1) and individuals (Level 3) cannot rely on reputation and trust alone. The individual may honor commitments expecting reciprocity; the corporation will defect whenever expected material gain exceeds expected material punishment. Sustainable cooperation requires contractual or regulatory constraints.

VIII. Computational Complexity

Computing Nash Equilibrium is PPAD-complete (Chen & Deng, 2006), meaning no polynomial-time algorithm is known and the problem is believed to require exponential time in the worst case. This raises the question: what is the computational complexity of finding AIE?

Conjecture (AIE Complexity). Computing AIE for finite multilevel games remains in PPAD when τ, μ, σ are polynomial-time computable.

Rationale: The AIE fixed point problem has the same structure as Nash: find s^* such that $s_i^* \in BR_i(s_{-i}^*; L)$ for all i. If U_i is computable in polynomial time given s and L, then the problem reduces to a standard Brouwer fixed point problem on the mixed strategy space, placing it in PPAD.

Overhead Analysis: Each payoff evaluation requires computing: $\tau(L(i))$ in $O(1)$ per player; $\Sigma\ \mu(L(i), L(j))$ in $O(n^2)$ for all pairs; $\sigma(L(i), s_{-i})$ in $O(n \cdot |S|)$ depending on opponent

strategies. Total overhead per evaluation: $O(n^2 \cdot |S|)$. For $n = 10$ players and $|S| = 100$ strategies, this adds 10^4 operations per evaluation.

Practical Implication: AIE is theoretically tractable (same complexity class as Nash) but practically harder (higher constants). Games with $n \leq 10$ and $|S| \leq 100$ are computable; larger games may require approximation methods.

IX. Empirical Predictions and Falsification

9.1 Distinguishing Predictions

Multilevel Game Theory generates predictions distinct from classical game theory:

Prediction 1 (Reputation Shock Response): Level 3 entities should reduce violations after reputation shocks even controlling for material consequences. Level 1 entities should show no response to pure reputation shocks. Empirical test: Compare corporate versus individual recidivism rates before and after media scandals, controlling for enforcement and revenue effects.

Prediction 2 (Contract Asymmetry): Contracts between Level 1 and Level 3 entities should exhibit more one-sided terms than contracts between entities at the same level. Empirical test: Compare optionality indices in corporate-consumer contracts versus corporate-corporate contracts versus individual-individual contracts.

Prediction 3 (Compliance Theater): Corporate compliance spending should correlate with enforcement visibility but not with actual wrongdoing reduction. Empirical test: Regress compliance investment on audit frequency and media coverage, controlling for violation rates.

Prediction 4 (Trust Game Asymmetry): In trust games, Level 3 trustees should return more than Level 1 trustees, even controlling for strategic considerations. Empirical test: Compare return rates by trustee type (individual versus corporate fiduciary) in experimental or field settings.

9.2 Falsification Criteria

The framework is falsifiable:

Falsification 1: If AIE predictions deviate from Nash predictions in a specific game, but empirical behavior matches Nash rather than AIE, the theory fails for that domain.

Falsification 2: If interventions designed to reduce intentionality asymmetry (e.g., requiring human decision-makers for corporate high-stakes choices) do not change equilibrium outcomes in the predicted direction, the mechanism fails.

Falsification 3: If cross-sectional variation in cooperation rates does not correlate with measured intentionality asymmetry across interaction types, the theory lacks explanatory power.

Falsification 4: If corporations exhibit shame responses (behavior change after reputation shocks controlling for material consequences) comparable to individuals, the Level 1 classification is incorrect.

9.3 Preliminary Empirical Evidence

While comprehensive empirical testing awaits future work, preliminary evidence from multiple domains supports the framework's core predictions. I summarize findings from corporate enforcement data that align with the theory's predictions about intentionality asymmetry.

Corporate Recidivism Patterns: Analysis of the Corporate Prosecution Registry (Garrett, 2014; updated through 2019) reveals that publicly traded corporations exhibit recidivism rates approximately 8 times higher than privately held firms (7.8% versus 0.9%). This pattern is consistent with the theory: public corporations, with greater separation between ownership and management and more proceduralized decision-making, operate closer to Level 1 than closely-held firms where individual owners bear direct consequences. Lund and Sarin (2021) document a steep rise in corporate recidivism from 2001-2018 and find that penalties represent 'inconsequential parking tickets' when scaled by firm assets, confirming that corporations process sanctions as costs rather than deterrents.

Enforcement Action Differentials: State securities enforcement data (NASAA, 2024) shows that firms face enforcement actions at twice the rate of individuals (46.3% versus 22.2% action rate from investigations). Federal data reveals an even starker pattern: corporations receive pre-trial diversion (Deferred Prosecution Agreements) at 30 times the rate of individual defendants (18% versus 0.6%), suggesting systematic recognition that corporate and individual defendants respond differently to identical legal frameworks.

DPA Breach Impunity: Public Citizen (2019) documents that 38 corporations account for 78 deferred prosecution agreements, with 32% breaching agreement terms and only 1.3% facing consequences for breaches. This pattern, where repeated violations and agreement

breaches produce minimal behavioral change, is precisely what the framework predicts for Level 1 entities immune to shame-based enforcement.

These patterns do not constitute definitive validation; confounding factors include firm size, enforcement resources, and detection probability differentials. However, the consistency of findings across environmental enforcement (EPA), securities regulation (SEC, NASAA), and criminal prosecution (DOJ) domains suggests the intentionality asymmetry mechanism operates broadly. A companion paper develops the full empirical analysis with appropriate controls and robustness checks.

X. Conclusion

This paper establishes the formal foundations of Multilevel Game Theory. Classical game theory assumes intentionality homogeneity: all players are Level 3 social reasoners capable of recursive belief attribution and susceptible to social enforcement. This assumption breaks down in modern strategic environments containing corporations (Level 1 optimizers), algorithms (Level 0.5 pattern matchers), and humans (Level 2-3 social reasoners). Multilevel Game Theory provides the mathematical framework for analyzing heterogeneous intentionality.

Three core results anchor the framework. The Existence Theorem establishes that AIE exists under standard conditions, ensuring the theory is well-posed. The Nash Equivalence Theorem establishes backward compatibility: when the classical homogeneity assumption holds, AIE reduces to Nash. The Gap Bounds Theorem quantifies the advantage from lower intentionality, providing empirically testable predictions.

The framework explains phenomena that puzzle classical game theory. Corporate criminal liability fails because it applies Level 3 deterrence mechanisms (shame, moral opprobrium) to Level 1 optimizers (cost-benefit calculators). AI ethics frameworks disappoint because they address Level 3 concerns to Level 0.5-1 systems. Terms of Service persistently exploit users because platforms (Level 1) are immune to the reciprocity norms that constrain Level 3 counterparties.

Multilevel Game Theory does not supplant classical game theory. It extends it. Just as relativity extends Newtonian mechanics, preserving Newtonian predictions at low velocities while correcting them at high velocities, Multilevel Game Theory preserves classical predictions under homogeneity while providing corrections under heterogeneity. The extension

is conservative: no classical result is overturned, only supplemented for cases where the homogeneity assumption fails.

The stakes are high. Regulatory regimes, contract doctrines, and international treaties are designed on the implicit assumption that all strategic agents are Level 3 social reasoners. When this assumption fails, designed institutions fail. The increasing prevalence of algorithmic decision-making, corporate optimization, and AI deployment makes intentionality heterogeneity the rule rather than the exception. Recognizing and formalizing intentionality asymmetry is the first step toward designing institutions that work in heterogeneous environments.

Future work extends in several directions. Empirical validation requires measuring intentionality levels systematically and testing the distinguishing predictions. Dynamic models where players choose intentionality levels endogenously would capture strategic delegation to algorithms. Applications to specific regulatory domains, from antitrust to financial regulation to AI governance, would demonstrate practical value. The formal apparatus is in place; the applications await.

References

Aumann, R. J. (1976). Agreeing to Disagree. The Annals of Statistics, 4(6), 1236-1239.

Berge, C. (1963). Topological Spaces. Dover Publications.

Camerer, C. F. (2003). Behavioral Game Theory: Experiments in Strategic Interaction. Princeton University Press.

Camerer, C. F., Ho, T.-H., & Chong, J.-K. (2004). A Cognitive Hierarchy Model of Games. Quarterly Journal of Economics, 119(3), 861-898.

Chen, X., & Deng, X. (2006). Settling the Complexity of Two-Player Nash Equilibrium. Proceedings of the 47th IEEE Symposium on Foundations of Computer Science, 261-272.

Chwe, M. S.-Y. (2001). Rational Ritual: Culture, Coordination, and Common Knowledge. Princeton University Press.

Dennett, D. C. (1987). The Intentional Stance. MIT Press.

Fudenberg, D., & Maskin, E. (1986). The Folk Theorem in Repeated Games with Discounting or with Incomplete Information. Econometrica, 54(3), 533-554.

Fudenberg, D., & Tirole, J. (1991). Game Theory. MIT Press.

Garrett, B. L. (2014). Too Big to Jail: How Prosecutors Compromise with Corporations. Harvard University Press.

Hansmann, H., & Kraakman, R. (2000). The Essential Role of Organizational Law. Yale Law Journal, 110(3), 387-440.

Hedden, T., & Zhang, J. (2002). What Do You Think I Think You Think? Strategic Reasoning in Matrix Games. Cognition, 85(1), 1-36.

Lewis, D. (1969). Convention: A Philosophical Study. Harvard University Press.

Lund, D. S., & Sarin, N. (2021). Corporate Crime and Punishment: An Empirical Study. Texas Law Review, 100(2), 285-342.

Marotta-Wurgler, F. (2007). What's in a Standard Form Contract? An Empirical Analysis of Software License Agreements. Journal of Empirical Legal Studies, 4(4), 677-713.

Maskin, E. (2008). Mechanism Design: How to Implement Social Goals. American Economic Review, 98(3), 567-576.

NASAA (2024). 2024 Enforcement Report. North American Securities Administrators Association.

Nash, J. (1950). Equilibrium Points in N-Person Games. Proceedings of the National Academy of Sciences, 36(1), 48-49.

Nash, J. (1951). Non-Cooperative Games. Annals of Mathematics, 54(2), 286-295.

Osborne, M. J., & Rubinstein, A. (1994). A Course in Game Theory. MIT Press.

Public Citizen (2019). Soft on Corporate Crime: Exposed: How the Trump DOJ Exposed Americans to More Corporate Malfeasance. Public Citizen.

Rubinstein, A. (1989). The Electronic Mail Game: Strategic Behavior Under 'Almost Common Knowledge.' American Economic Review, 79(3), 385-391.

Russell, S. (2019). Human Compatible: Artificial Intelligence and the Problem of Control. Viking.

Schelling, T. C. (1960). The Strategy of Conflict. Harvard University Press.

Selten, R. (1965). Spieltheoretische Behandlung eines Oligopolmodells mit Nachfrageträgheit. Zeitschrift für die gesamte Staatswissenschaft, 121, 301-324.

Von Neumann, J., & Morgenstern, O. (1944). Theory of Games and Economic Behavior. Princeton University Press.

Weibull, J. W. (1995). Evolutionary Game Theory. MIT Press.